

Assessing Superhuman Speed as a Gamified Reward in a Virtual Reality Bike Exergame

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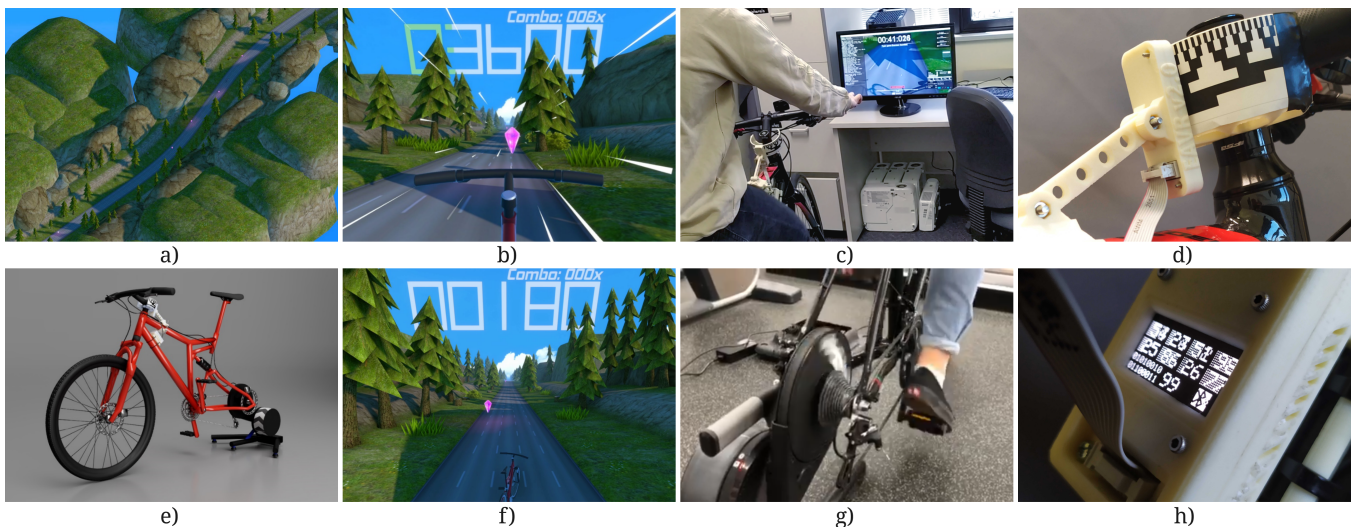


Figure 1: a) birds-eye view of the track and scenery, b) optical-flow effect during speed boost, c) a user controlling the bike with the steering device and the user's view inside the system, d) custom rotary encoder for steering device, e) render of the complete bike apparatus, f) third-person steering method, g) KICKR trainer attached to the bike, h) output display for steering device.

ABSTRACT

This research presents a Virtual Reality (VR) bike exergame system consisting of a stationed road bike, four novel virtual steering methods, two in-game superhuman speed reward ideas, and a pedalling system simulating real-life resistance. A study is conducted to understand how virtual speed can be used as a reward to encourage enjoyment in physical activities. The result suggests that adding reward-based superhuman speed positively impacts user enjoyment. Adding steering control and increasing speed potentially do not inhibit greater motion sickness symptoms using well-aligned physical-to-virtual input and output representations.

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CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Interaction design*; *Interaction devices*; • **Software and its engineering** → **Interactive games**.

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1 INTRODUCTION

The Australian guidelines for physical activity reported on Oct 2020, recommend adults do at least 150 minutes of moderate to vigorous physical activity per week. 55% of Australian adults did not meet these physical activity guidelines¹. Insufficient physical activities

¹<https://www.aihw.gov.au/reports/risk-factors/insufficient-physical-activity/contents/insufficient-physical-activity>

are a major risk factor for chronic conditions, therefore, encouraging people to increase physical activities through new methods becomes imperative. VR allows users to be fully immersed in their environments and contains enough detail, complexity and responsiveness to provide compelling experiences. It has been shown to improve the user's ability to exercise, reduce tiredness and increase relaxation after exercise [18], and improve the enjoyment and performance of high-intensity interval training [6]. Gamification is defined as the process of taking something people find mundane and boring and making it more interesting and fun through the implementation of interactive game mechanics. Exercise games or exergaming is one form of gamification applied to physical activities. Exergames have already shown benefits over traditional exercise. Not only can it improve physical performance, but it also boosts enjoyment and dissociation [7]. VR gives us a platform to create exergames that virtually push the human body past what it is normally capable of, giving people superhuman abilities [13]. User accomplishments are a big part of gamification, especially for exergames. Motivational rewards such as power-ups and high scores have already been shown to increase physical activity [21] and result in greater health benefits for the user [24]. However, the gap between perceived motion in VR and the actual physical movements in real life can be misaligned, making speed a complicated variable to represent by providing users with a sense of superhuman ability and not causing motion sickness during this activity.

In this study, we investigate an unexplored area of research examining reward-based superhuman speed in a VR exergame by exploring how manipulation of perceptive elements in a virtual environment can affect the enjoyment of physical activity. A VR exergame system with four customised steering methods and pedalling resistance is presented with two types of gamified rewards integrated to increase virtual speed. A pilot and a main study are conducted, aiming to understand how perceived speed can affect the user's response in an exertion gaming platform and how best speed can be applied in game design to improve health benefits (in the form of motion sickness). The goal is to create a more beneficial, enjoyable, and engaging VR bicycle experience for the end user.

1.1 Main contributions of research

First, one of the first studies used a bike exergame system that simulates real-life steering and pedalling controls in VR. Second, it presents a prototype consisting of a VR-integrated bike with four novel steering methods to avoid motion sickness, real-life resistance capabilities simulating uphill and downhill movements, and two in-game reward methods, including a superhuman speed boost. Third, it provides insights collected from a user study to examine how reward-based superhuman speed improves the enjoyment of physical activity using VR.

2 BACKGROUND

2.1 Exercise Bikes and Superhuman Sports in VR

The use of VR bike systems is becoming more prevalent in both tech and health science domains. This is partly due to the simplicity of stationary bike exercises, and the bike's power output accuracy as a

measure of the work participants do. Previous studies suggested that participants have noticed the speed differences [22] and enjoyed the open-world in-game experience of riding a VR bike [9]. Performance shows a positive effect when video feedback is associated with music [14] and when a ghost player motivates participants to work harder towards enjoyment [5]. However, they only assessed speed as a deception technique rather than a measure to understand how differentiation affects the psychological response to exercise enjoyment.

Superhuman sports are a combination of physical activity, design and technology. Increasing the virtual speed of a bicycle to make the user feel as though they are more powerful and capable would fall under the domain of superhuman sports. By utilizing new technologies, including VR and augmented reality (AR), superhuman sports can provide users with activities going beyond the limits of natural human ability [1, 13, 15, 16]. Studies have shown that AR/VR superhuman sports encourage physical activity by providing users with an immersive experience [10] and haptic feedback in the form of resistance through the bike pedals leads to greater potential to increase enjoyment [19]. However, using immersive technology alone limits user experience to the advancement of the current state, making a fast-paced sports game less natural to participate fully. As far as we know, no research has been done to use steering as an input to control VR bikes. Combining sensors, haptic resistance, and a customised steering system with a VR bike system could be an alternative to overcome technical limitations and still provide a realistic experience.

2.2 Gamification and Perception in VR

Gamifying an exercise system is the process of turning it into something with repetitive value and increased enjoyment. Reward, an important aspect of gamification, delivers a user a sense of accomplishment; when incorporated correctly, it can be the deciding factor in enjoyment. For example, linking game rewards to physical exercise can improve motivation and increase physical activity [21]. In a VR bike exergame, enhanced users' capabilities, a competitive high-scoring system, and visual stimuli to superhuman levels are deemed helpful in gamification [17].

Optical flow is the pattern of objects in motion within our field of view. For example, visualised as lines of motion, optical flow helps us determine the speed at which we travel and the speed an object travels versus its distance away. In VR, optical flow plays a large role in the perception of speed. For example, on a bicycle, optical flow moves quickest in our peripheral and slowest in our central field of view. This means a scene's field of view could play a role in how we virtually perceive motion. Studies suggest that in VR, people perceive their speed to be slower than what it actually is [2], resulting in a negative effect on enjoyment as people feel they work harder than their speed entails [4]. This counters the goal of making the exergame enjoyable. Researchers believed riding a bike involves movement and gaze in a straight-ahead direction, so slower perceived speed was caused by the lower field of view (FOV) angle [2, 23]. VR could also increase the chance of motion sickness caused by a sensory disconnect between the perceived and visualised speed [20], leading to lower user enjoyment.

2.3 Hypotheses & Research Question (RQ)

To summarise, previous research has shown the benefit of using VR to create bike exergame to encourage physical exercise and enjoyment. However, it is unknown if using increased speed as a gamified reward exerts a psychological response to improve health benefits and how speed should be represented through a VR interface to align perceived and visualised forms to increase enjoyment. In this study, we hypothesise that a well-designed speed increase visual stimulus as a reward inside a VR bicycle exergame will increase the enjoyment of physical exercise as well as bring health benefits.

RQ1: Does superhuman speed as a gamification reward raise the enjoyment in VR bicycle exergame? **RQ2:** Does adding physical steering input cause greater motion sickness in VR bike exercise?

3 SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Overview

We present an exergame system (Figure 1) that integrates a stationed physical road bike with a VR head-mounted display (HMD). The system is customised to provide the following features: first, physical bike riding experience. The road bike is fixed with sensors to station to a fanning zone (wind simulation) to mimic a real-life biking experience. Second, the user can physically steer and pedal a bike as input and is aligned in a 1:1 ratio on a virtual track with varying realistic uphill and downhill sections and physics. Third, a reward system consisting of a scoreboard and superhuman speed boost to encourage continuous physical activities. A video demonstrating the system can be found here ².

3.2 Customised Bike Control System

The customised bike controls, including physical bike handlebars, pedals, and sensors, acts as input for the VR steering system. Pedalling and steering are means of locomotion inside VR. It moves the user forward and grounds them in the virtual environment. In our system, the users control the bike's forward movement by pedalling their legs and steering left to right by turning the handlebars with their arms like game controllers. They also receive output in the form of pedal resistance. Hence the pedalling force varies depending on whether they are going uphill or downhill, leading to a congruence between their effort and the perceived steepness of the current track. For example, realistic inertia occurs on downhill slopes and the resistance from travelling uphill is enough to slow the user down in a realistic way. The resistance is determined by the track's gradient in the VR world and communicated to the bike's wahoo KICKR trainer.

Motion sickness is caused by the difference between expected steering forces and felt steering forces. The stationary bike prevented the user from leaning, a large component of steering a non-stationary bike. This produced an inaccuracy in the user's proprioception whilst performing realistic turns on the bike in VR, resulting in over-corrected steering, disorientation and motion sickness. To reduce this, we created and tested four methods of VR steering: 1) Accurate steering, this meant if the user steered the real bike to the left, the virtual bike accurately followed the action and turned as a normal bike would (in a curved path); 2) Fixed steering,

in which the virtual bike laterally translates from left to right whilst maintaining an always forward direction. For example, when the user steers to the left at a small angle, the VR bike moves to the left horizontally (along the x-axis only) until it hits the left bound; 3) Third-person steering has the users' perspective follow behind and above the bike and steers like the accurate steering method but via a birds-eye-view; 4) Percentage steering extends fixed steering by replacing the free hand steering to a mapping between the user's x-axis and their handlebar rotation degree corresponding to the track's width. For example, the handlebar position at 0° means the bike is at the centre of the track or at 45° means it should go to the rightmost.

3.3 VR Bike Interface

Our system uses track pieces as segments of an entire bike track in VR. As the user rides along the track, more track pieces are appended onto the end to form one continuous path. The track piece's slope percentage, height, and length are parameterised and can be adjusted before run-time to create a custom or completely randomised track. Tracks can be infinitely long or end after a certain pre-specified time or length, in which case a finish line is generated as the final track piece.

The scenery is procedurally generated as each track piece is created. Scenery objects like trees, rocks, and bushes are placed pseudo-randomly between specific vertex positions on the track piece mesh. Their local position and chance to spawn are completely random but within a bound to avoid obstructing the user's trajectory along the track. All the scenery objects use a custom resource pooling system we developed, whereby scenery objects are loaded into memory when the application starts, retrieved when needed for a new track piece, and then returned when that track piece is destroyed. This prevents unwanted halting or stalling when creating track pieces, making for a smoother user experience.

Inside VR, the user's virtual headset is positioned at a known offset to the bike's virtual handlebars to align them with the physical handlebars. Collectable crystals are placed at fixed intervals along the bike track's vertical direction (z-axis) and randomly along the horizontal direction (x-axis). When a user collects a crystal, it shatters into multiple pieces, sending those pieces toward a large floating scoreboard in the sky.

3.4 In-Game Reward System

Rewards are an important part of gaming, and choosing the correct reward for users' achievements could encourage them to continue playing. This translates well into our exergaming scenario to encourage rigorous physical activities. In our system, a facility-type reward [17] is presented as an increase in speed when users collect a group of "crystals" (game items) along the bike track. Their scores will be added in real-time, followed by a reward with a speed boost.

An active scoreboard component shows the user their current score and multiplier bonus. The scoreboard is at a far distance from the user to avoid distraction. It moves with the user subtly as they travel forward to prevent them from riding past it. To reinforce the reward system, we added a particle effect to the user's peripheral view in the VR screen that replicates lines of optical flow, a technique used in racing games to emphasize users' movements

²https://www.youtube.com/watch?v=ObJ_zpfWCZA

at high speed. This stylized choice made speed more noticeable to the user and matched the overall art style of the game assets.

Our system presents two methods for conveying speed rewards: cumulative increase and speed boost. The first increases the user's base speed by a fixed number with each subsequent crystal collected. For example, the speed will cumulatively increase by 5% when the crystal is collected. The second gives users a 2x faster speed boost every 100 meters for 15 seconds when the crystal is collected. For example, if users pedal hard and fast to reach the next crystal before the current 15 seconds elapse, they will result in a constant 2x speed. Each exercise session is set to 10 minutes.

3.5 Implementation

The system (Figure 1) was built with a medium-sized female Merida Speeder 400 road bike, HTC Vive Pro VR HMD, Bluetooth Arduino steering device, and Wahoo KICKR resistance trainer. The virtual environment and bike system were developed inside the Unity game engine (2017.1.1f1). The rideable track was generated in three phases. The bike track extruded a polygonal mesh along a bezier curve with predefined control points in the first phase. In the second phase grass, bushes, trees, and rocks were procedurally placed along the bike track with random scale, rotation, and position parameters. The final phase generated the physics colliders, trigger boxes, and invisible walls to ensure the virtual bike stayed aligned with the track. A custom rotary encoding device was developed to afford steering capabilities as a new control input method. The device incorporated an 8-track Gray code cylinder (Figure 1: upper right) with an accuracy of 1.4 degrees across 256 positions, an array of eight QRE1113 infrared reflective sensors for each track in the Gray code disk, and an Adafruit Huzzah ESP 32 BlueTooth Arduino device (Figure 1: lower right) for sending data to the system. A steering angle was calculated by interpolating between the maximum left and right angles of the physical handlebars based on the Gray code position. A smoothing algorithm was then applied to the angle to reduce any jitter caused by the BlueTooth connection. The virtual handlebars were rotated to match the calculated steering angle. A wahoo KICKR resistance trainer was connected to the rear axle of the road bike to record the user's pedalling speed and apply electromagnetic resistance to the pedals. The maximum simultaneous power output of the KICKR is 2200W and has an accuracy of $\pm 2\%$ [25]. The KICKR was connected to the system via the ANT+ communication technology. Through this connection, the user's speed was extrapolated into forward momentum, and the physical pedal resistance was programmatically increased or decreased during virtual ascents and descents, respectively.

4 USER STUDY

4.1 Pilot Testing

An iterative process was used to ensure that the exercise task was functional, rewarding, and enjoyable. Four VR experts participated in the pilot evaluation. Four steering methods were tested to avoid motion sickness before settling on the fixed steering technique, which was later presented in the formal study. The fixed steering method was preferred by everyone, although some disorientation remained. The controllability of the virtual bike increased drastically, and the user's tendency to over-correct their steering was

greatly reduced. For accurate and percentage steering, users found them produce too much oversteering or disorientation, making the game difficult to control. The third-person method felt disconnected from the real-person experience, making the rewards feel detached from their original immersive purpose. We also tested varying game lengths and determined that 10 minutes was sufficient enough to provide a short rigorous exercise. The two reward methods were also piloted. The cumulative method was too minimal to notice to be a reward at the initial stage. In contrast, the speed boost method was constantly noticeable to have an impactful effect on the participants. Therefore we settled with the speed boost for the main study.

4.2 Participants

Twelve participants (10 male and 2 female) aged between 20 to 35 years old ($M(SD) = 28(4.3)$) were recruited from a research university. All participants had ridden a bike before, with three riding weekly, one riding monthly, and eight riding yearly. They all had used a VR system within a year, with seven using it more than once a month and five several times a year.

4.3 Study Design

The study is a within-subject in two levels (1x2). The conditions are: "with speed boost" (**W**) and "without speed boost" (**O**). During the W condition, when a crystal was collected, the participant's virtual speed was increased by 2x coupled with a visual effect depicting optical flow speed lines. The O condition has no increase in virtual speed upon collecting a crystal as the baseline. Both conditions equally present the rest of the features and their corresponding visual feedback, including tracks, sceneries, the scoreboard, crystals, etc. The conditions are counterbalanced. Each participant rides along an identical bike track randomly generated in both conditions to control consistency and avoid bias.

5 MEASURES

Pre-study questionnaire: Simulator Sickness questionnaire (SSQ) [12] is included to understand users' symptoms of motion sickness before the task as baseline data. **Post-condition questionnaires:** consists of two questionnaires. SSQ and Physical Activity Enjoyment Scale (PACES) [11]. SSQ is used to understand the participant's motion sickness level compared to their pre-study score and against each condition. PACES compares users' enjoyment of the exercise performed in each condition in a 7-point Likert scale format. Question 17 of the PACES questionnaire ("*It's very refreshing*") is removed as it's deemed irrelevant to our system. **Post-study questionnaire:** asks for users' preference of the conditions followed by an open-ended question for their general feedback. This measure provides extra qualitative data users might have.

5.1 Task and Procedure

The study room consisted of a customised medium-sized female road bike, a VR HMD, a fan, and a computer. The bike was stationed at the fanning zone to create a wind effect while the researcher sat opposite the room, controlling the computer. The seat height was adjustable, and two different seat styles were available, one soft and one hard. Participants were asked to perform two ten-minute

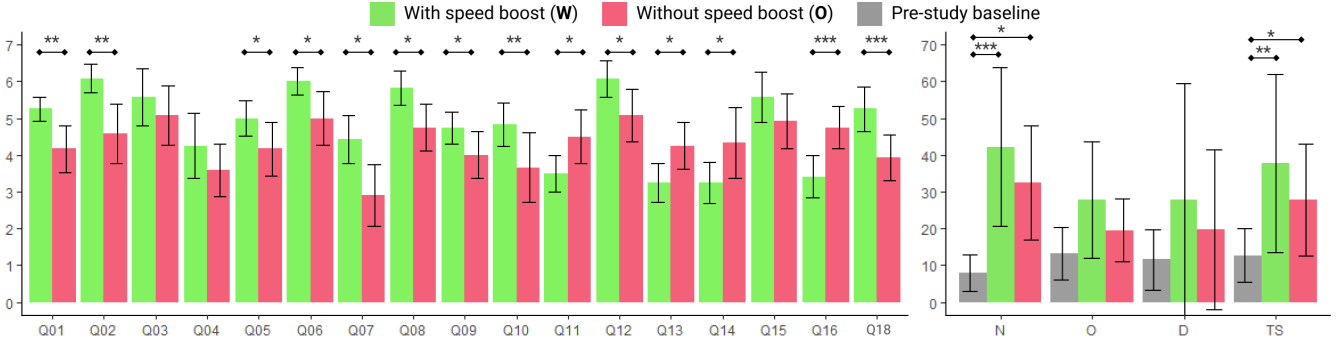


Figure 2: PACES (left) and SSQ (right) in subscales of (N, O, D, TS) between "With speed boost" (W) and "without speed boost" (O).

biking exercises using our VR prototype. During both conditions, participants were informed when they hit the five-minute and thirty seconds time stamps. They were expected to play the game at their own pace, collecting crystals placed along the bike path. Participants were asked to review the information sheet, sign a consent form and answer a demographic questionnaire before the study. They were then given a brief introduction followed by a two-minute system familiarisation on a plain track with no crystals or scoreboard to ride a VR bike. Subsequently, participants completed two VR bike exercise conditions, each lasting ten minutes (10x2=20 mins in total). A pre-study questionnaire, post-condition questionnaires (2 in total), and a post-study questionnaire were collected. Participants were allowed to ride the bicycle at any pace they felt comfortable and were reimbursed with a confectionary for their involvement.

6 RESULTS

6.1 Physical Enjoyment

T-tests were applied to the total response score adopting [8] method. **W** ($M \pm SD = 97.3 \pm 13.3$) had a significantly better score than **O** ($M \pm SD = 80 \pm 20.8$) condition ($t(p) = 2.43 (< .05)$). Wilcoxon Signed Rank tests were applied to each 7-Likert scale PACES question (17 in total in Figure 2: left) after reversing a given number of questions following [11] (in Appendix). The result indicated that **W** received significantly higher ratings than condition **O** as a more enjoyable (Q1: $V(p) = 36(.01)$), interesting (Q2: $V(p) = 52(.01)$), absorbed (Q5: $V(p) = 21(.03)$), fun (Q6: $V(p) = 28(.02)$), energizing (Q7: $V(p) = 34(.02)$), happy (Q8: $V(p) = 21(.03)$), pleasant (Q9: $V(p) = 26(.04)$), feel good physically (Q10: $V(p) = 36(.01)$), invigorating (Q11: $V(p) = 26(.04)$), not frustrating (Q12: $V(p) = 26(.05)$), gratifying (Q13: $V(p) = 41(.02)$), exhilarating (Q14: $V(p) = 26(.04)$), a sense of accomplishment (Q16: $V(p) = 63(< .01)$), and nothing else I would rather be doing (Q18: $V(p) = 45(< .01)$) activity. For the post-study preference question "Which condition did you prefer?", all twelve participants preferred the **W** condition.

6.2 Motion sickness

Following the method adopted in [12] (in Appendix), SSQ was grouped into three subscales and a total score [3] (Figure 2: right): Nausea (N), Oculomotor (O), Disorientation (D), and the total score (TS). Since user data did not follow a normal distribution, Wilcoxon

Signed Rank Tests were applied. To compare against the baseline before the study, we found statistical significance in both conditions **W** and **O** in Nausea ($W:V(p) = 0(< .01)$; $O:V(p) = 1(.01)$) and the total score ($W:V(p) = 3(.01)$; $O:V(p) = 7.5(.04)$), but not in Oculomotor or Disorientation. Therefore, a certain degree of motion sickness exists in VR regardless of conditions, but it is relatively minimal in our system.

7 DISCUSSION AND LIMITATION

The significance found in the total PACES score indicated that participants had a greater enjoyable response to the "with speed boost" (**W**) condition (**RQ1**). Every participant supported this result by choosing **W** as the preferred condition in the post-study questionnaire. Open-ended qualitative feedback further suggested that participants believed the "without speed boost" (**O**) condition to be "boring" (P3) and **W** felt "a lot more realistic" (P7). Participants also commented that they felt like they rode "a greater distance" (P5) than they thought was personally achievable. This was also supported by PACES Q16, where participants reported a significantly greater sense of accomplishment when they had an increased speed over without. We proposed and piloted four novel steering methods to reduce motion sickness, and the results suggested that the method we adopted (fixed steering) still posed some motion sickness for participants. This research lends insights into the effects of the chosen physical input and virtual output method and opens opportunities for further exploration into steering methodologies and how the affect motion sickness (**RQ2**).

A limiting factor of our system was the absence of the user's hands and arms inside VR, which could cause disassociation between the virtual and physical worlds reducing overall immersion. The fan used to simulate wind was set to a fixed rate. Instead, a variable rate through synchronizing it with the user's forward momentum could have produced a more immersive experience and further emphasised the speed-boosting effects. We designed a steering method to overcome the limitations of using a stationary bike that could not lean, however, future research could explore ways of incorporating a leaning or motorised platform to allow the user to lean more naturally on the stationary bike.

8 CONCLUSION

This research presents a VR bike exergame system that incorporates a customised steering and pedalling system coupled with in-game superhuman speed as a reward to increase the enjoyment of physical exercise. A pilot study evaluates four novel steering and two reward methods. A formal study is conducted to understand physical activity enjoyment and motion sickness between a superhuman speed boost and a no-reward condition. Results suggested that adding reward-based superhuman speed positively impacts user enjoyment during physical activity. It also shows that increasing speed inhibits some motion sickness symptoms if the user's movements inside of VR do not align with their movements on the stationary bike. This research has demonstrated a valuable system for innovative ideas to help encourage physical activity amongst our population through a means that is more rewarding and enjoyable to the end user using VR. Future work, such as motion sickness caused by the alignment between the steering method, superhuman sports, and VR representation, can be further investigated as to why some had a greater adverse effect than others.

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9 APPENDIX

	None	Slight	Moderate	Severe
General discomfort	0	1	2	3
Fatigue	0	1	2	3
Headache	0	1	2	3
Eye strain	0	1	2	3
Difficulty focusing	0	1	2	3
Salivation increasing	0	1	2	3
Sweating	0	1	2	3
Nausea	0	1	2	3
Difficulty concentrating	0	1	2	3
Fullness of the head	0	1	2	3
Blurred vision	0	1	2	3
Dizziness with eyes open	0	1	2	3
Dizziness with eyes closed	0	1	2	3
Vertigo*	0	1	2	3
Stomach awareness**	0	1	2	3
Burping	0	1	2	3

* Vertigo is experienced as loss of orientation with respect to vertical upright.

** Stomach awareness is usually used to indicate a feeling of discomfort which is just short of nausea.

Figure 3: Simulator Sickness Questionnaire

Appendix							
Physical Activity Enjoyment Scale							
Please rate how you feel <i>at the moment</i> about the physical activity you have been doing.							
*	1	2	3	4	5	6	7
I enjoy it							I hate it
	1	2	3	4	5	6	7
I feel bored							I feel interested
	1	2	3	4	5	6	7
I dislike it							I like it
*	1	2	3	4	5	6	7
I find it pleasurable							I find it unpleasurable
*	1	2	3	4	5	6	7
I am very absorbed in this activity							I am not at all absorbed in this activity
	1	2	3	4	5	6	7
It's no fun at all							It's a lot of fun
*	1	2	3	4	5	6	7
I find it energizing							I find it tiring
	1	2	3	4	5	6	7
It makes me depressed							It makes me happy
*	1	2	3	4	5	6	7
It's very pleasant							It's very unpleasant
*	1	2	3	4	5	6	7
I feel good physically while doing it							I feel bad physically while doing it
*	1	2	3	4	5	6	7
It's very invigorating							It's not at all invigorating
	1	2	3	4	5	6	7
I am very frustrated by it							I am not at all frustrated by it
*	1	2	3	4	5	6	7
It's very gratifying							It's not at all gratifying
*	1	2	3	4	5	6	7
It's very exhilarating							It's not at all exhilarating
	1	2	3	4	5	6	7
It's not at all stimulating							It's very stimulating
*	1	2	3	4	5	6	7
It gives me a strong sense of accomplishment							It does not give me any sense of accomplishment at all
*	1	2	3	4	5	6	7
It's very refreshing							It's not at all refreshing
	1	2	3	4	5	6	7
I felt as though I would rather be doing something else							I felt as though there was nothing else I would rather be doing

*Item is reversed scored (i.e., 1=7, 2=6, . . . 6=2, 7=1).

Figure 4: Physical Activity Enjoyment Scale